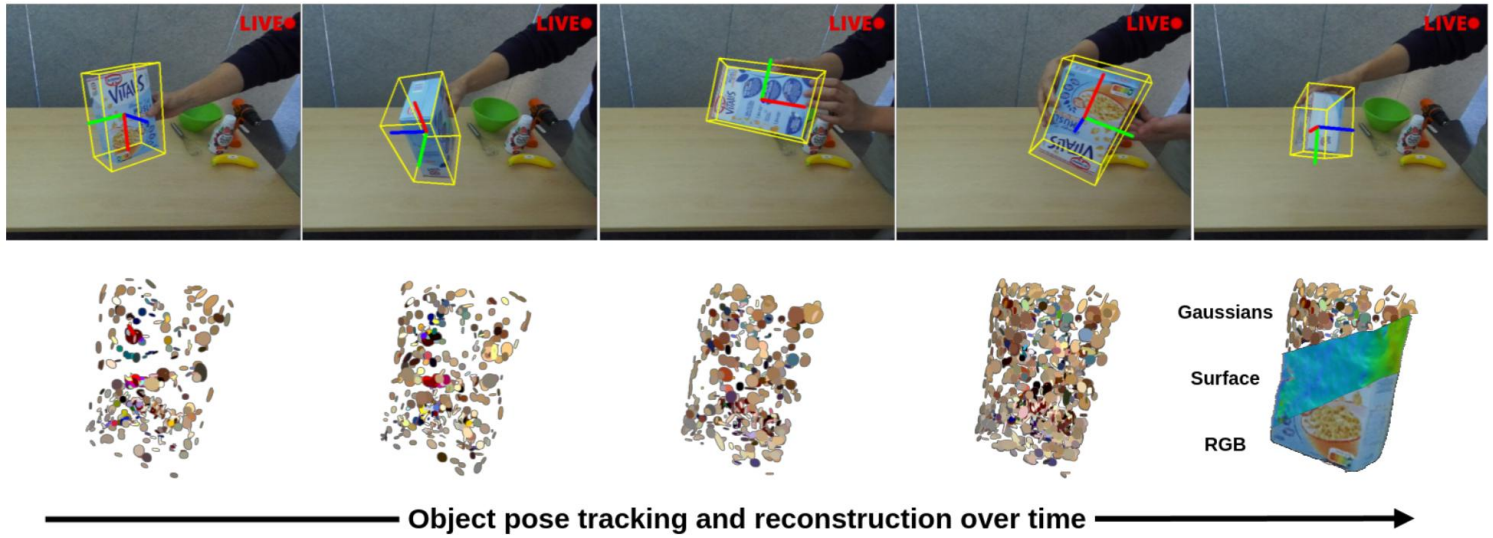
6DOPE-GS: Online 6D Object Pose Estimation using Gaussian Splatting

(ICCV 2025)



한국과학기술연구원(KIST)

CVIPL 학생연구원 **김연욱**

한국과학기술연구원
Korea Institute of Science and Technology

Contents

- **Preliminaries**
- Overview
- Introduction
- Method
- Experiment
- Conclusion

Preliminaries

Object Pose Estimation:

- a fundamental computer vision problem
- aims to **estimate the pose of an object** in a given image relative to the camera that captured the image.

Preliminaries

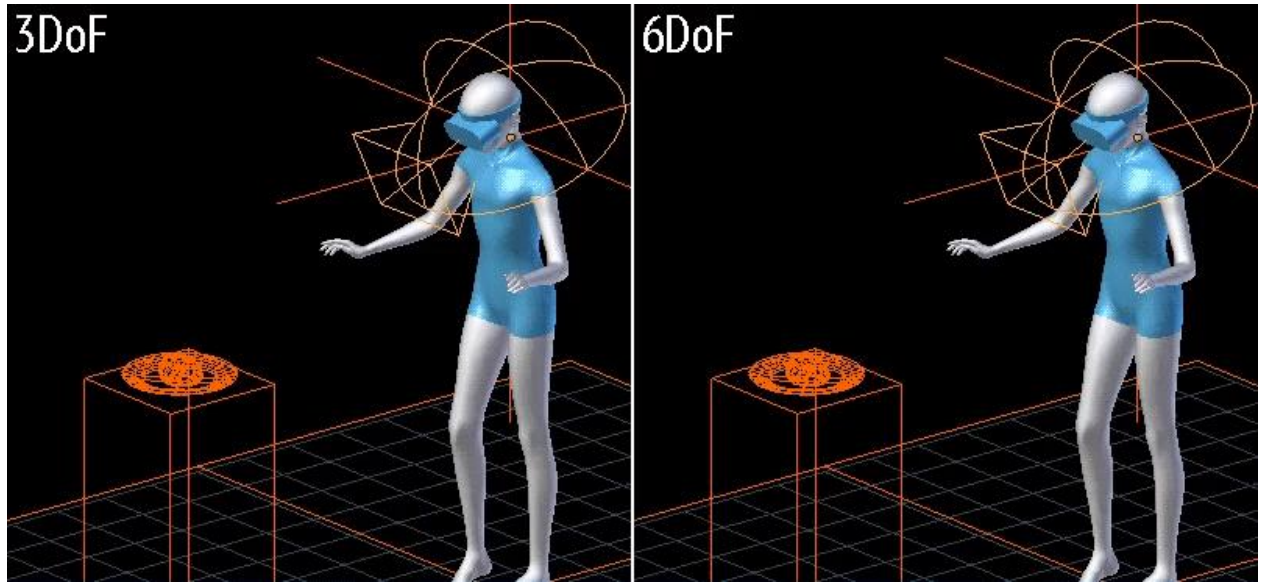
Application of OPE

- Autonomous driving
- Augmented reality
- Robotic manipulation
- Etc…

Preliminaries

Varying Degrees of Freedom (DoF)

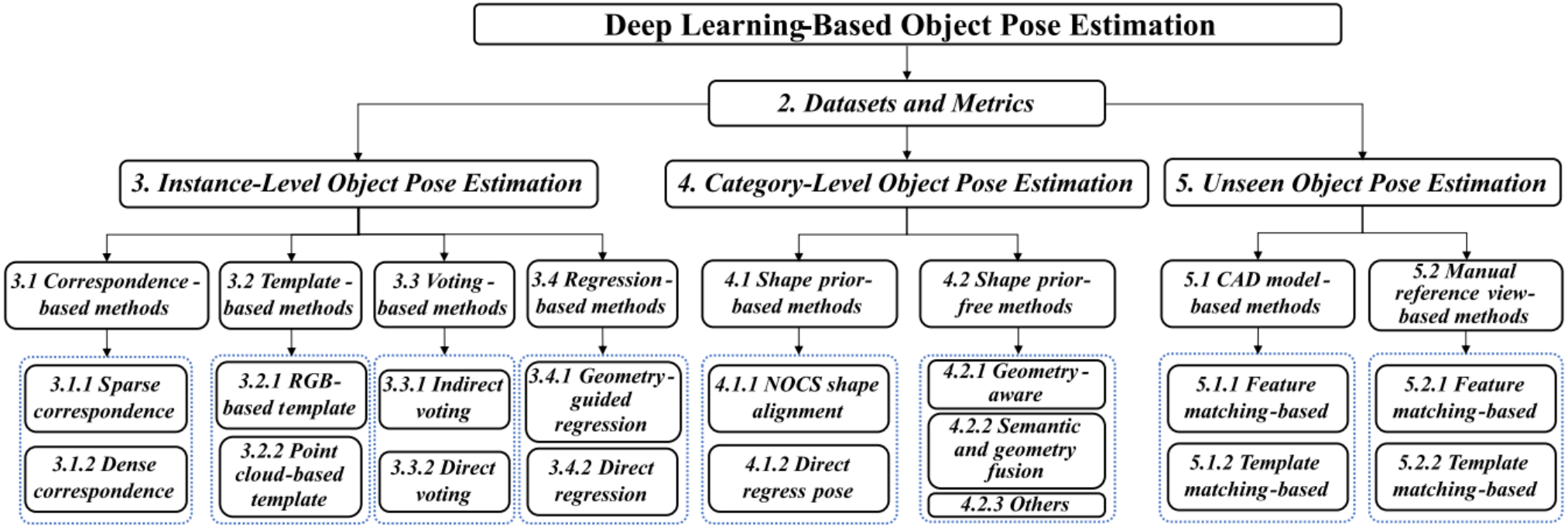
- **3DoF** : 3D rotation
- **6DoF** : 3D rotation + 3D translation
- **9DoF** : 3D rotation + 3D translation + 3D size of the object



Preliminaries

Taxonomy of OPE

- **Instance-Level -** estimating the pose of a **specific, exact object instance**
- **Category-Level -** estimating the pose of **various distinct objects** that belong to a **class**.
- **Unseen Object Pose Estimation -** estimating the pose of completely novel objects



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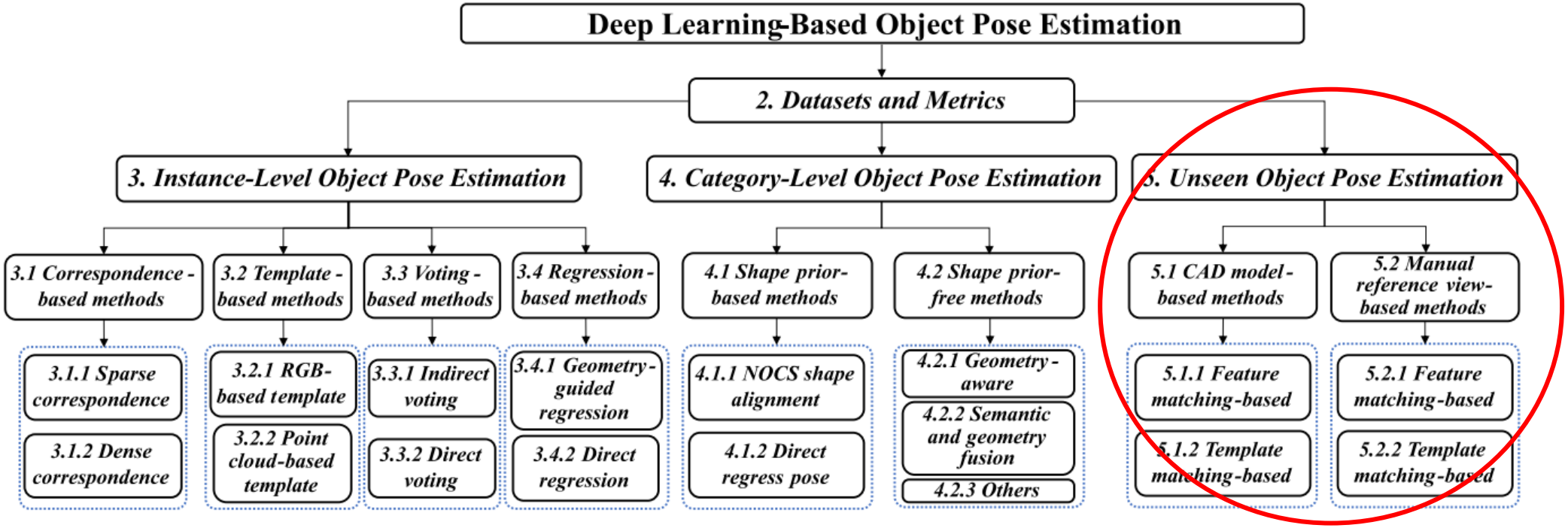
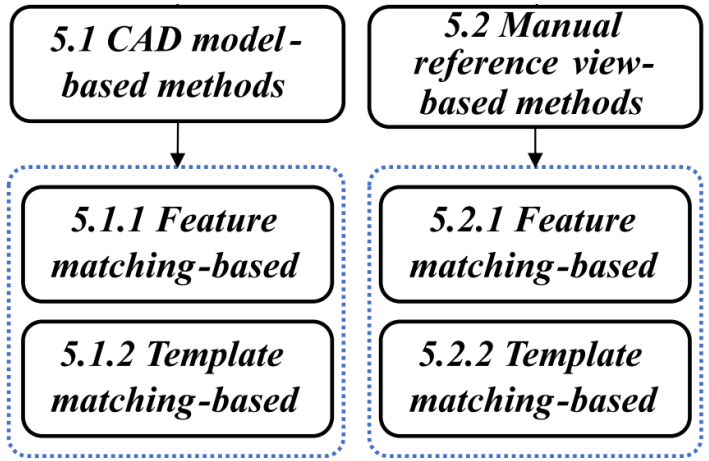


Preliminaries

Unseen Object Pose Estimation

- **Model-based**
  - 사전에 주어지는 object의 **3D CAD 모델**(.ply, .obj 등) 을 **직접 활용**하는 방식
  - **입력 이미지**에서 추출한 특징과 **3D CAD 모델**의 기하학적 정보를 매칭하여 **pose 추정**

- **Model-free**
  - 사전 **3D CAD 모델 없이** 객체의 자세를 추론하는 방식
  - 최근에는 **소수의 Reference 이미지**로 NeRF나 3DGS 같은 Neural Radient Field를 **즉석에서 학습**시키고, 이 **신경망이 객체의 3D 모델 역할을 대신**한다.
  - 이후 **rendering한 이미지**와 **실제 입력 이미지**를 비교하여 **오차를 줄이며 pose 추청**



Preliminaries

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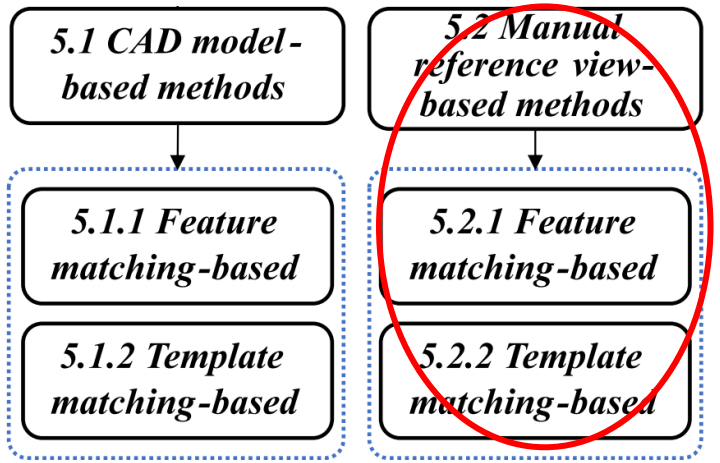


Similar to SLAM

6DOPE-GS

OverView

6DOPE-GS:

jointly optimizes **object poses** from observed keyframes and **reconstructs** a 3D object model on the fly using incremental **2D Gaussian Splatting**.

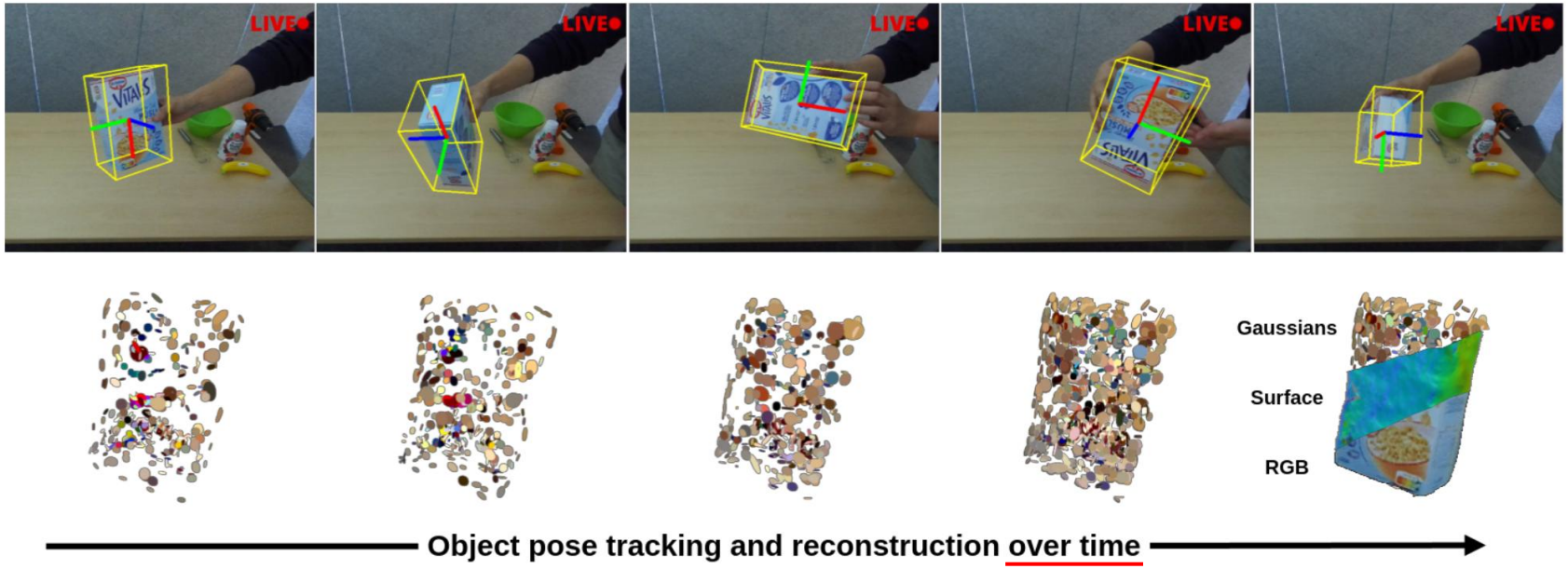


Figure 1. Demonstrating live object pose tracking and reconstruction of a test object in the real-world using 6DOPE-GS: a novel method for joint 6D object pose estimation and reconstruction using Gaussian Splatting. **Top:** 6D pose estimates of the object over time, **Bottom:** Example reconstruction over time with 2D Gaussian disks used to render the surface and appearance of the object. Our method enables live pose tracking and Gaussian Splat reconstruction of dynamic objects at 3.5Hz.

Introduction

Background

- 정확한 6D Pose Estimation 및 3D Reconstruction은 Robot Manipulation과 AR 환경에서 필수적인 기술이다.
- model-based 6D OPE는 유망한 결과를 내놓았지만,
- model-free 방식은 **Rendering**과 **inferencing**에 필요한 **높은 연산량**에 의해 한계에 부딪혀왔다.

Zero-Shot Object Pose Estimation

- Nevertheless, there has been **exciting progress in zero-shot**, **model-free** methods over the **past few years**.
  (Like BundleSDF, FoundationPose…)

Introduction

BundleSDF:

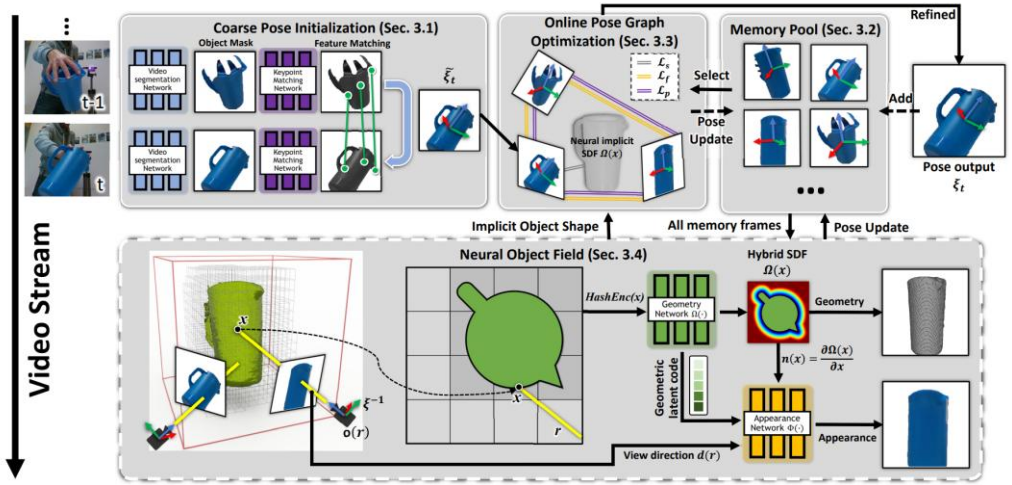
jointly optimize a "**neural object field**" and the **object poses** by learning a 3D Signed Distance Field**(SDF)** representation while **concurrently** running a **global pose graph optimization**.

Limitations:

Despite reporting near real-time pose optimization capabilities (~10Hz),
the **neural object field training** is **far** from real-time, (~0.4Hz).

→ leverage 2D Gaussian Splatting

- Rasterization 기반으로 Real-time rendering
- 2DGS ensures accurate geometric alignment
  by converting each Gaussian into a disk-like surfel
  (similar to SDF, but faster)



Method

Architecture Overview

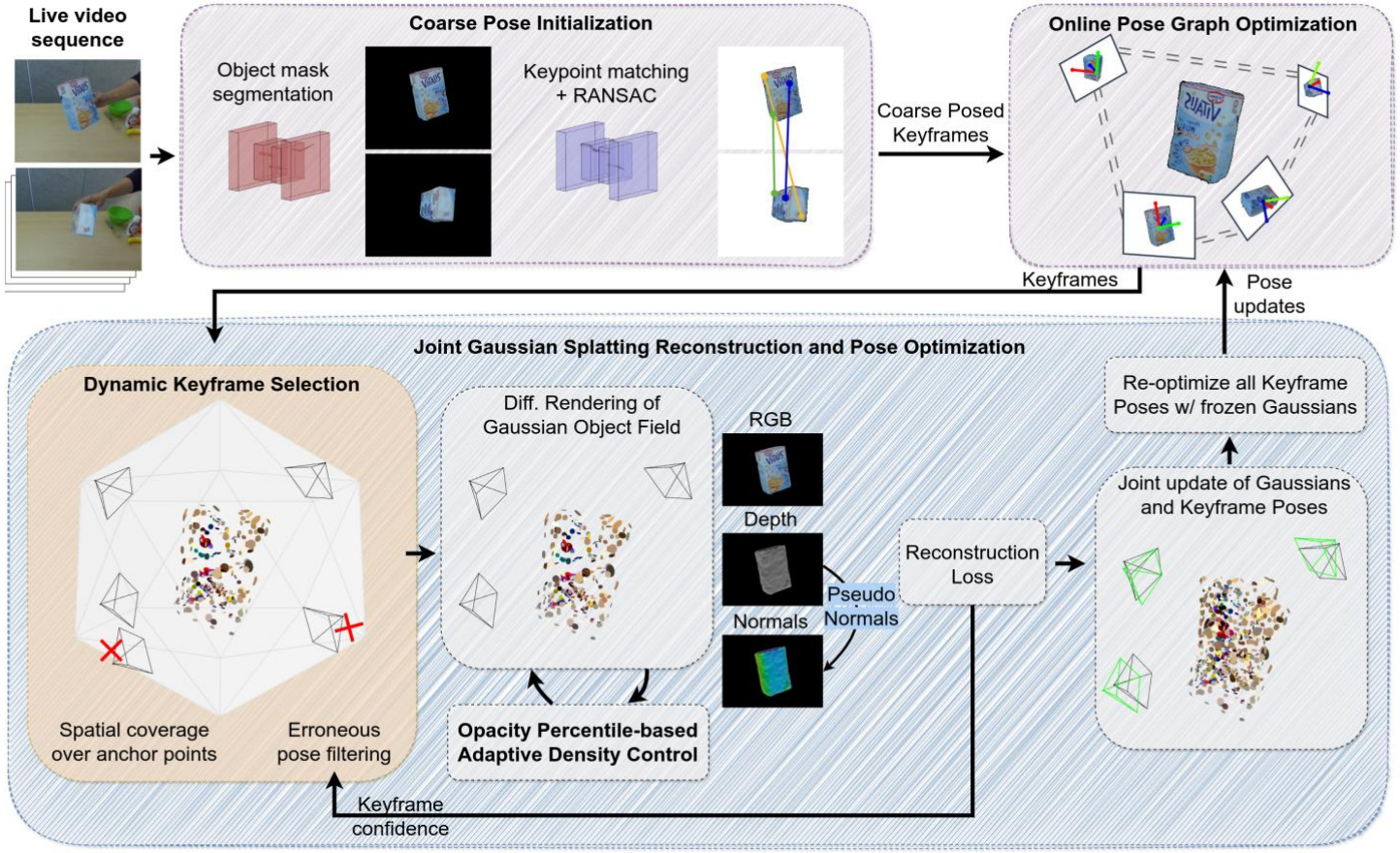
Task: Online 6D Object Pose Tracking

(비디오가 재생되는 동안 실시간으로 물체의 위치와 자세를 추적)

Input: RGB-D, object mask

Output:

6D Pose, 2D gaussian representation



Method

Architecture Overview

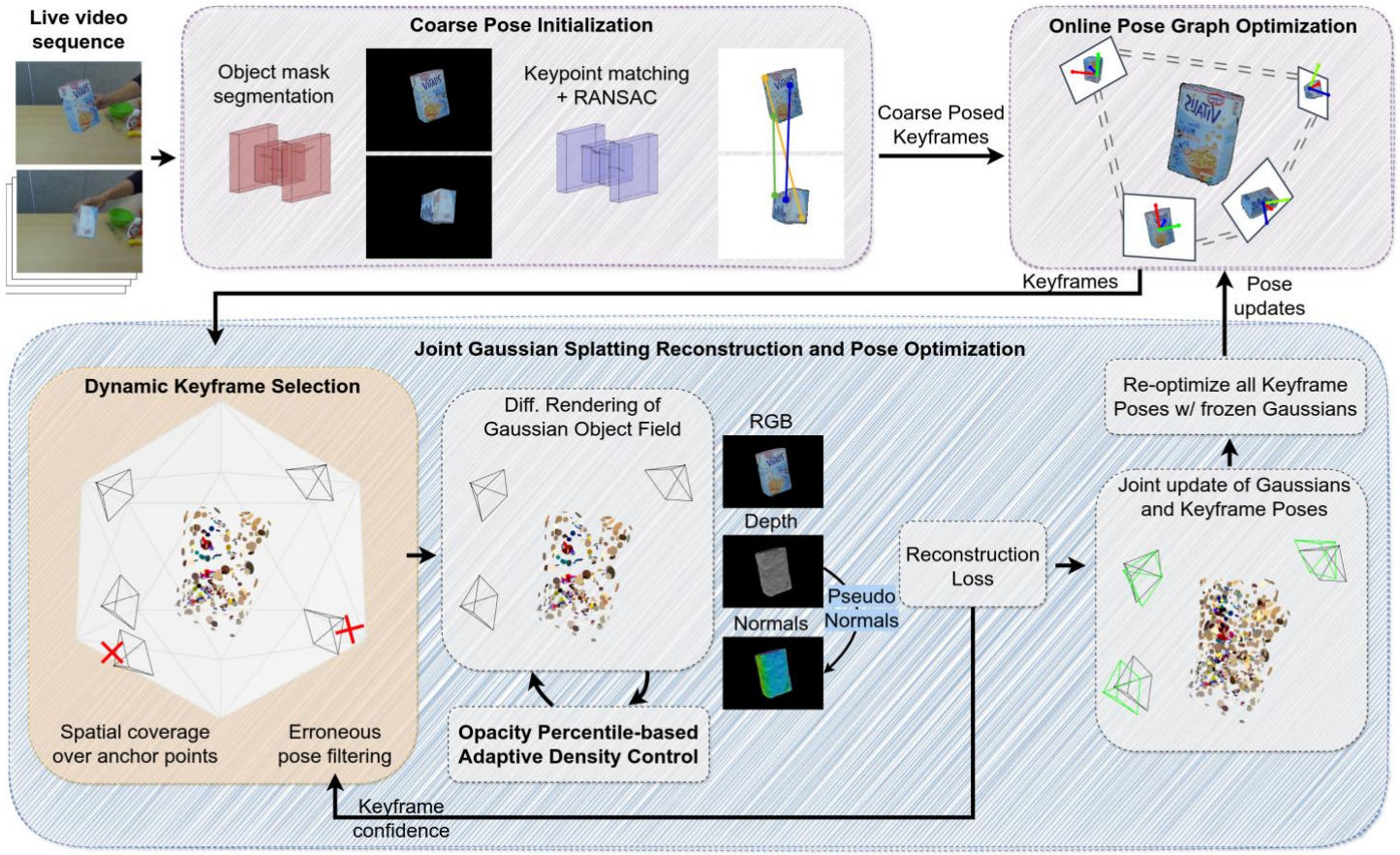
3.1. Coarse Pose Initialization

3.2. Gaussian Object Field

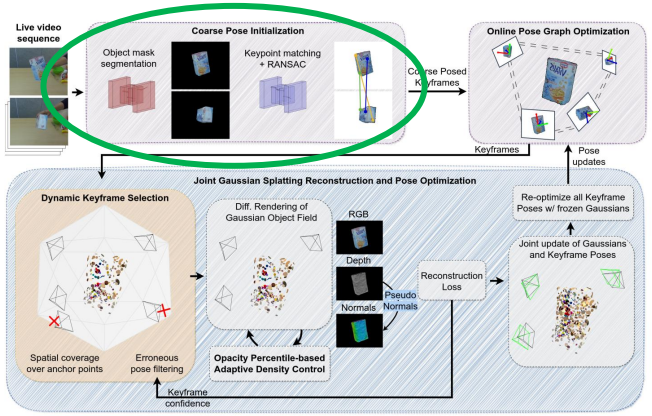
3.3. Dynamic Keyframe Selection for Gaussian Splatting Optimization

3.4. Opacity Percentile-based Adaptive Density Control

3.5. Online Pose Graph Optimization



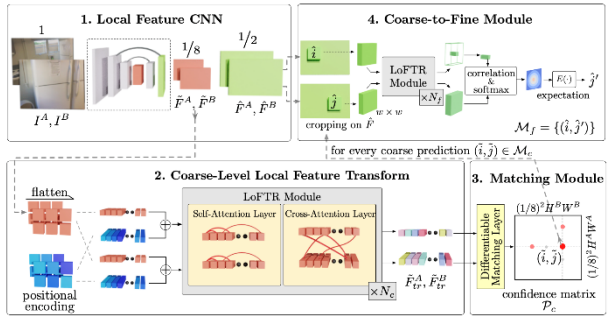
Method



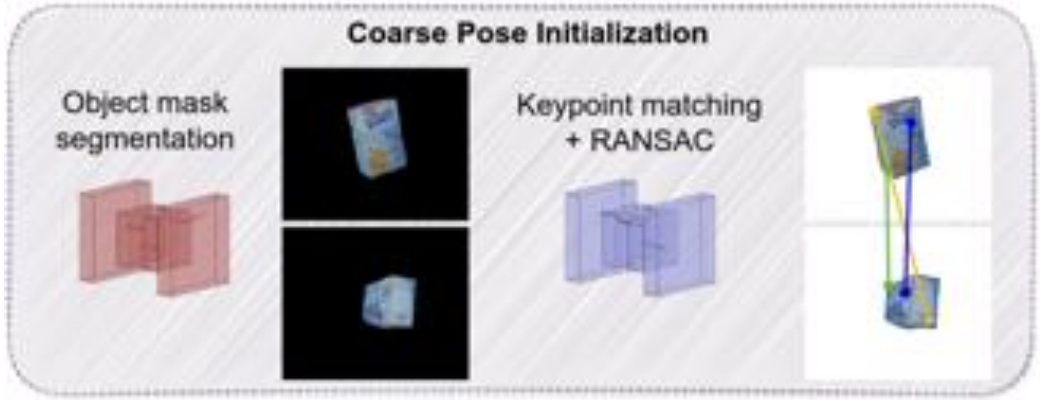
3.1. Coarse Pose Initialization

- First, use **SAM2** for facilitating effective **segmentation** and **tracking** of the **object** in question.
- Then use **LoFTR**(a transformer-based dense feature matcher) to estimate **feature** point **correspondences.**
- Compute **a coarse pose estimate(camera pose)** between pairs of **RGB-D** frames with **nonlinear least-squares** optimization in a **RANSAC** fashion.

  (Depth 정보를 결합한 3D-2D 매칭(Perspective-n-Point, PnP) 방식)



Loftr



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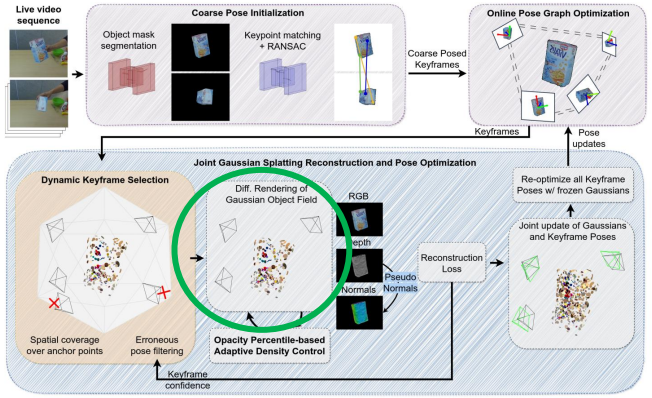


3.2. Gaussian Object Field

build a Gaussian Object Field using **2D Gaussian Splatting (2DGS)**
Because 2DGS ensures accurate **geometric alignment** by converting each Gaussian into a **disk-like surfel**. (similar to SDF)

3D Gaussian Primitives

- $\boldsymbol{\mu}$: 3D centroid (Mean)
- $\boldsymbol{\Sigma}$: Covariance matrix (decomposed into scaling matrix **S**, rotation matrix **R**)
- $\boldsymbol{c}$: Color (Spherical harmonic coefficients)
- α : Opacity

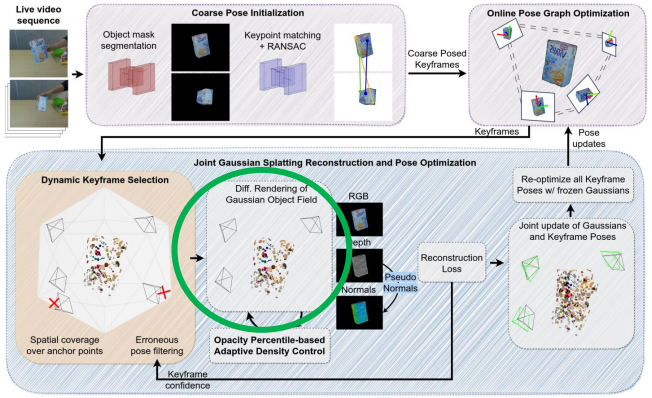
$$\{\mathbf{g}_k = (\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \boldsymbol{\alpha}_k, \mathbf{S}_k)\}_k^K$$

$$\boldsymbol{\Sigma} = \boldsymbol{R}\boldsymbol{S}\boldsymbol{S}^\top\boldsymbol{R}^\top$$

$$\boldsymbol{S} = diag([s_x, s_y, s_z])$$

$$\boldsymbol{R} \in SO(3)$$

Method



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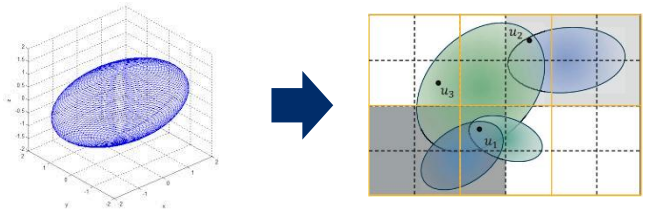
$$\boldsymbol{R} \in SO(3)$$

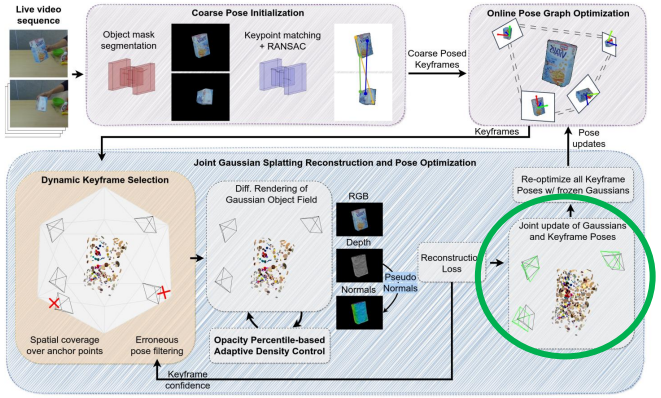
In 2DGS, **z-component** of the **scaling** matrix is set to **zero** $\boldsymbol{S} = diag([s_u, s_v, 0])$ for each Gaussian.

3D Gaussian ➔ 2D oriented planar Gaussian disks (with two principal axes $\mathbf{t_u}$ and $\mathbf{t_v}$)



- Normal : $\boldsymbol{t}_w = \boldsymbol{t}_u \times \boldsymbol{t}_v$, Rotation Matrix : $\boldsymbol{R} = [\boldsymbol{t}_u, \boldsymbol{t}_v, \boldsymbol{t}_w]$

Method

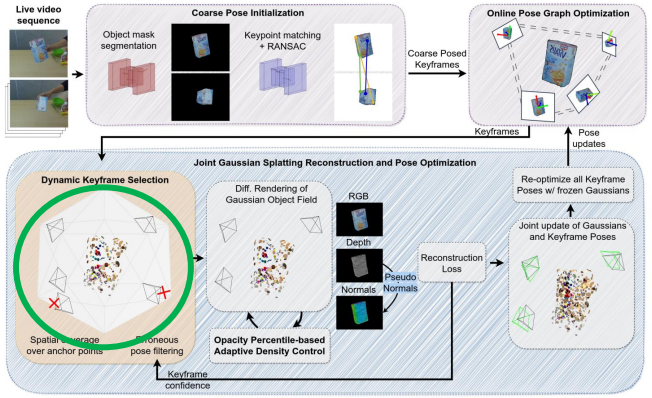


3.3. Dynamic Keyframe Selection for Gaussian Splatting Optimization

- **Once** we obtain a **coarse pose initialization** of **keyframes**,

  we aim to **construct** a **2DGS model** of the object.
- However, as the number of **keyframes increases**, the computational **cost** also **increases**.
- A **dynamic keyframe selection** approach **to mitigate** these issues.

Method

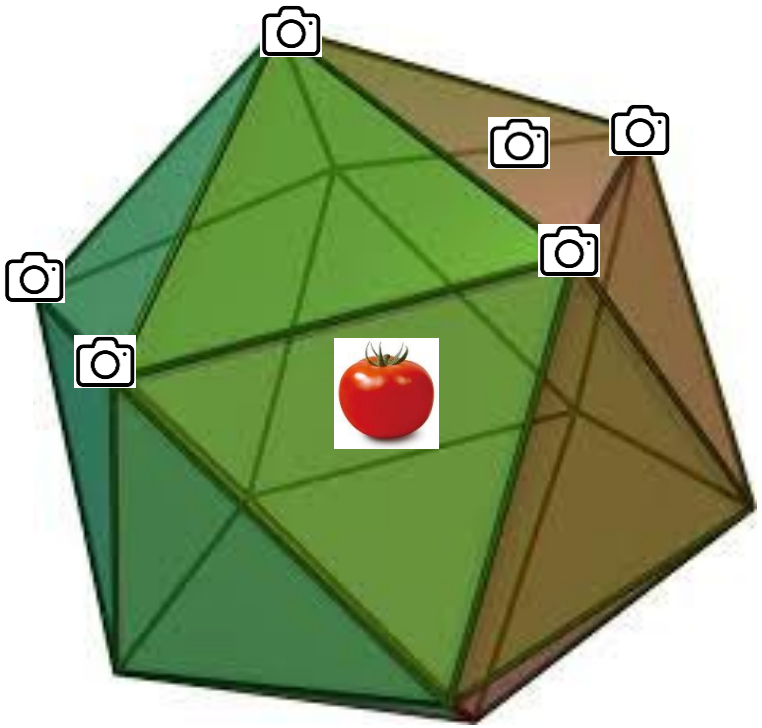


Dynamic Keyframe Selection

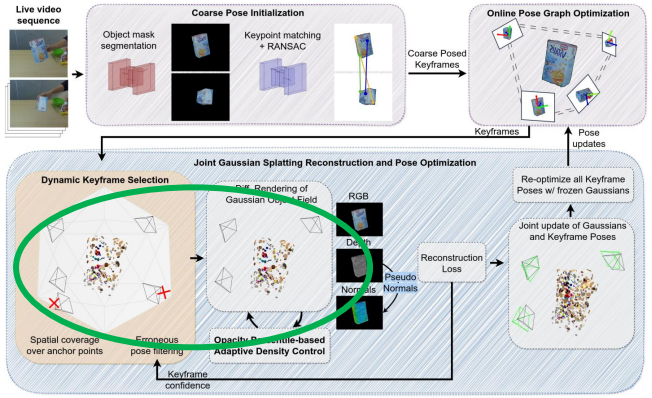
- First, establish a series of **“anchor points”** at **varying resolution** levels, using the **vertices** and **face** centers of an **icosahedron**(20면체).

- Then, **cluster** the **initial coarse keyframe pose estimates** around these anchor points along the icosahedron.

- **Select** the **keyframe** with the **largest object mask** in the cluster of each anchor point **for efficiency**.

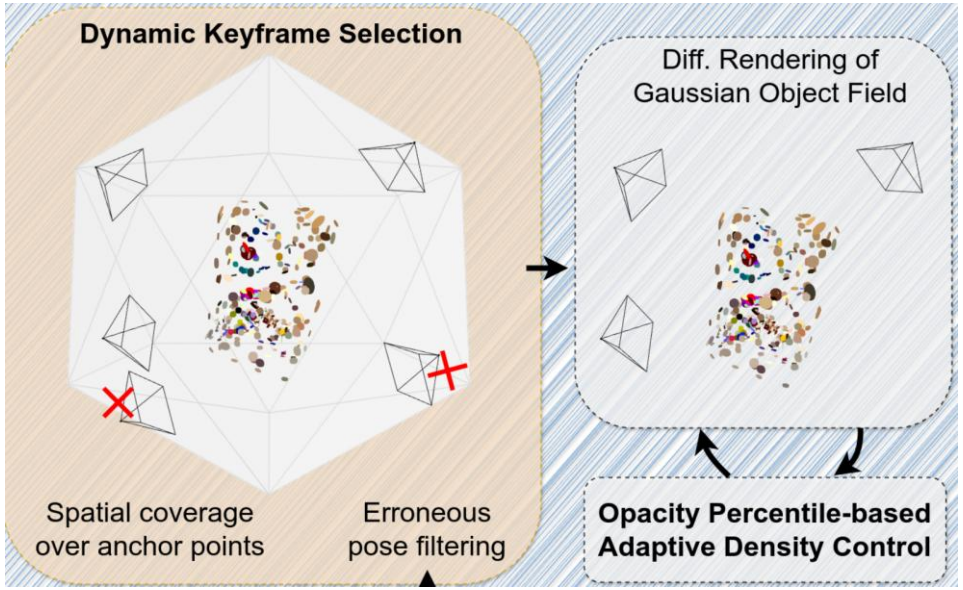


Method



Dynamic Keyframe Selection

- While **jointly optimizing the 2D Gaussians** and the **selected keyframe poses**, we **further remove outliers** with **erroneous pose estimates based on** the **reconstruction error** (obtained during the 2D Gaussian optimization.)
- Specifically, estimate the **median absolute deviation (MAD)** of the reconstruction loss (**MAD** is roubst to outlier.) (평균은 outlier에 약하다.)



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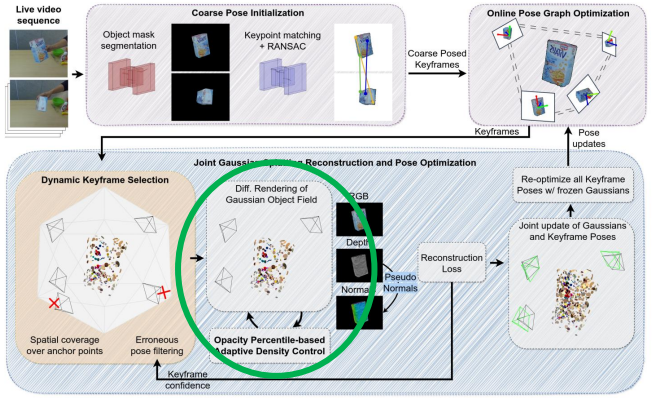
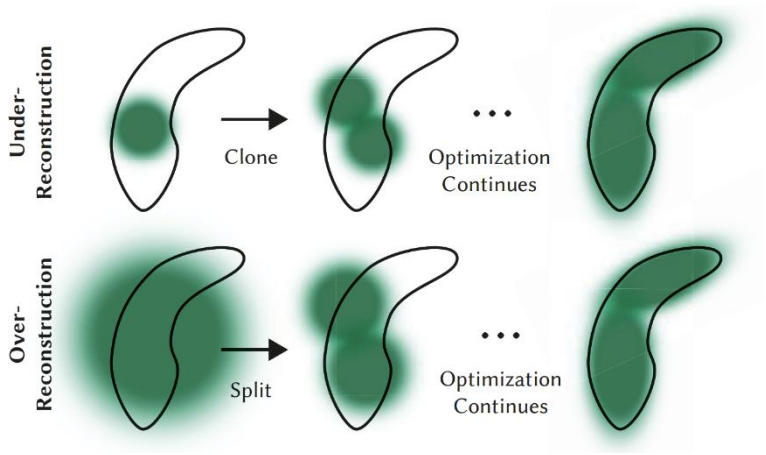


3.4. Opacity Percentile-based Adaptive Density Control

perform **pruning** and **densification** During the **optimization of the Gaussian Object Field**

Vanilla Adaptive Density Control

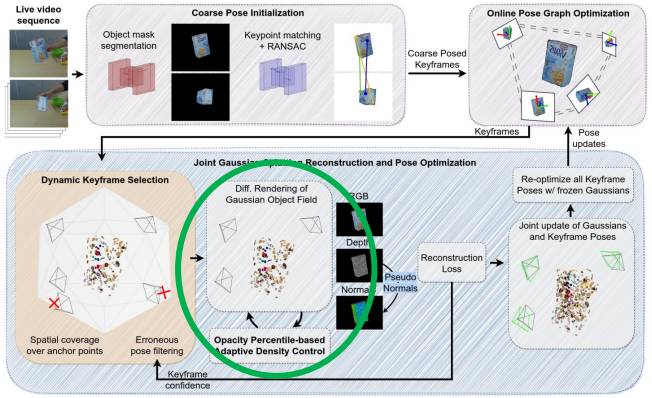
- 크게 이동해야 하는 가우시안들을 Split하거나 Clone하여 빈 공간을 채우고,
- 투명도가 너무 낮아진 가우시안은 주기적으로 삭제.



➔ It demands considerable engineering work

Method

3.4. Opacity Percentile-based Adaptive Density Control

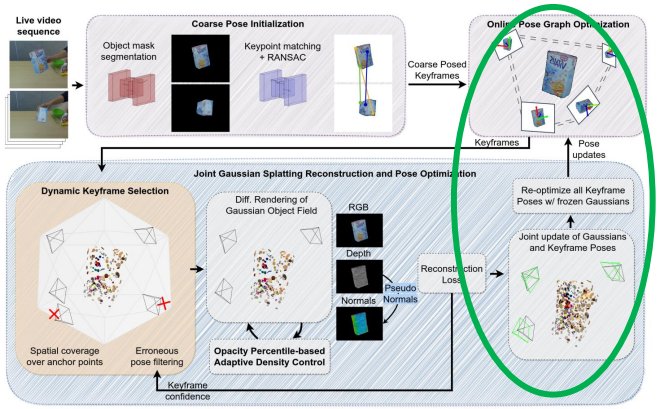


Since we have an **object-centric focus**,
limit the **scale** of the Gaussians and instead use a **percentile-based pruning strategy**.

After a **fixed number** of optimization steps,
prune the Gaussians with **opacity** in the **bottom 5th** percentile **until** the **opacity** of the **95th** percentile of the Gaussian particles **exceeds** a given **threshold**.

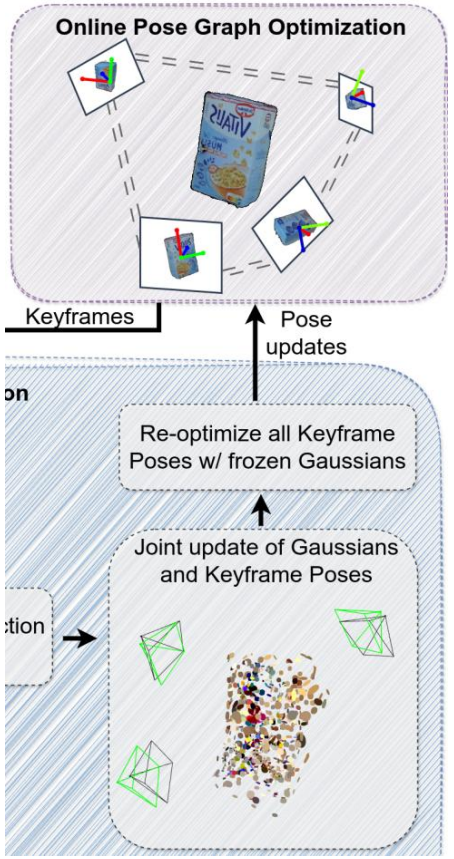
➔ 가우시안 수의 안정화
➔ 학습의 안정화
➔ 불필요한 가우시안 제거
➔ (But, still non-differential)

Method

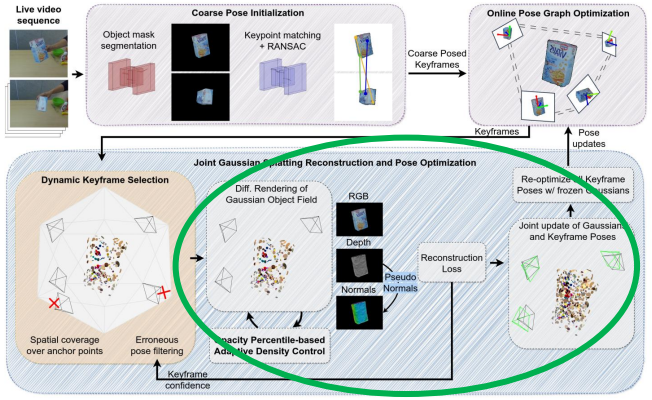


3.5. Online Pose Graph Optimization

When we **receive** the **updated poses** for the **keyframes from Gaussian Object Field**, we establish a **global object-centric coordinate system** and a **keyframe memory pool**, which **stores key correspondences**.



Method



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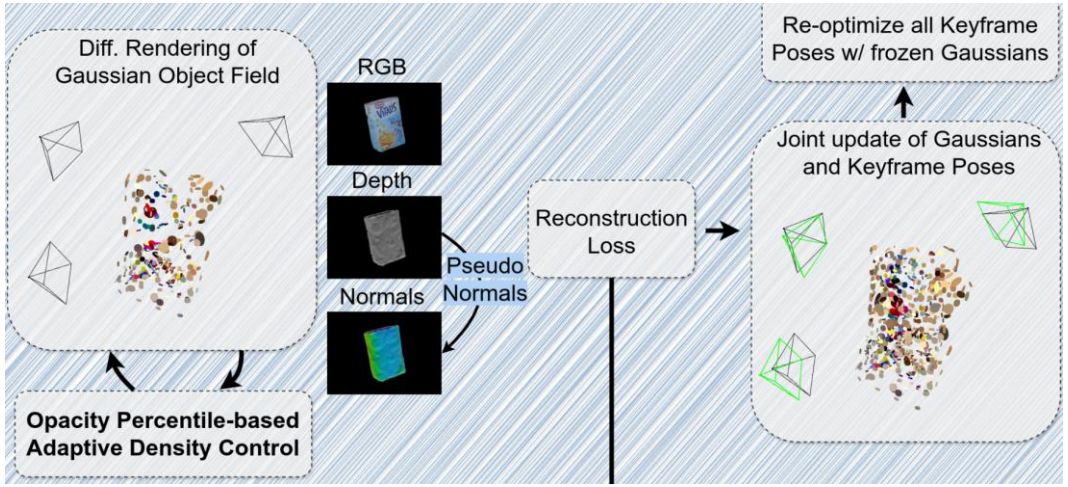
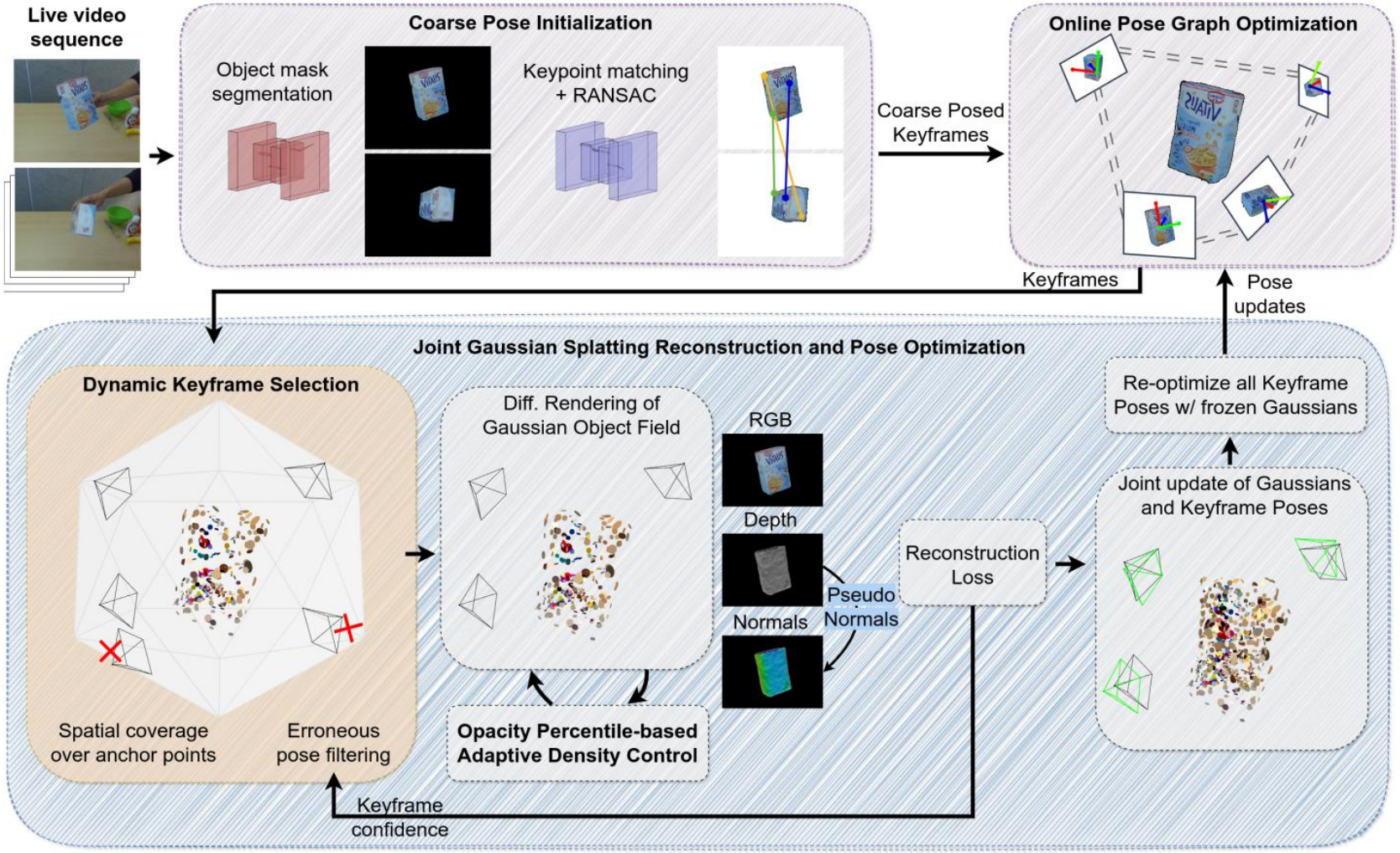


- Generate a **point-normal map** for each frame in **pool**.
- **Compute** the **dot-product** of the **normals** with the **camera-ray direction** of the **new frame**, to **assess visibility**.
- If the **visibility ratio** of the new incoming frame **exceeds** a defined **threshold**, frames are **selected**.

  ➔ These frames are **simultaneously** used as the backbone for **3D reconstruction** and **global pose graph optimization**.

Method

Over-view



Experiment

Evaluate the methods based on **three** key metrics:

- 6-DoF object **pose tracking** accuracy
- 3D **reconstruction** accuracy
- computational **efficiency**.

Datasets

YCBInEOAT: dataset for testing performance in a **'relatively calm** environment'

HO3D: dataset for testing performance in the 'most **rugged**(occludes and shakes) environment'

YCBInEOAT

HO3D

Experiment

Quantitative Comparison

Method	Pose		Reconstruction	Efficiency
	ADD-S (%) ↑	ADD (%) ↑	CD (cm) ↓	ATPF(s) ↓
NICE-SLAM [78]	23.41	12.70	6.13	n.a.
SDF-2-SDF [55]	28.20	14.04	2.61	n.a.
DROID-SLAM [61]	46.39	34.68	4.63	n.a.
MaskFusion [52]	41.88	35.07	2.34	n.a.
MonoGS(RGB-D) [35]	20.16	15.32	2.43	0.29
BundleTrack [69]	92.54	84.91	-	**0.21**
BundleSDF [71]	92.82	84.28	0.53	0.82
6DOPE-GS	**93.79**	**87.82**	**0.15**	0.22

Table 1. Comparison on the YCBInEOAT Dataset. ADD and ADD-S are reported as AUC percentages (0 to 0.1m), and reconstruction accuracy is measured by Chamfer loss. ATPF is the average processing time per frame (n.a. indicates unavailable data).

YCBInEOAT

Method	Pose		Reconstruction	Efficiency
	ADD-S (%) ↑	ADD (%) ↑	CD (cm) ↓	ATPF(s) ↓
NICE-SLAM [78]	22.29	8.97	52.57	n.a.
SDF-2-SDF [55]	35.88	16.08	9.65	n.a.
KinectFusion [40]	25.81	16.54	15.49	n.a.
DROID-SLAM [61]	64.64	33.36	30.84	n.a.
MonoGS(RGB-D) [35]	2.81	1.82	22.09	0.36
BundleTrack [69]	93.96	77.75	-	0.29
BundleSDF [71]	94.86	**89.56**	0.58	2.10
6DOPE-GS	**95.07**	84.33	**0.41**	**0.24**

Table 2. Comparison on the HO3D Dataset. ADD and ADD-S are reported as AUC percentages (0 to 0.1m), and reconstruction accuracy is measured by Chamfer loss. ATPF is the average processing time per frame (n.a. indicates unavailable data).

HO3D

Experiment

Quantitative Comparison

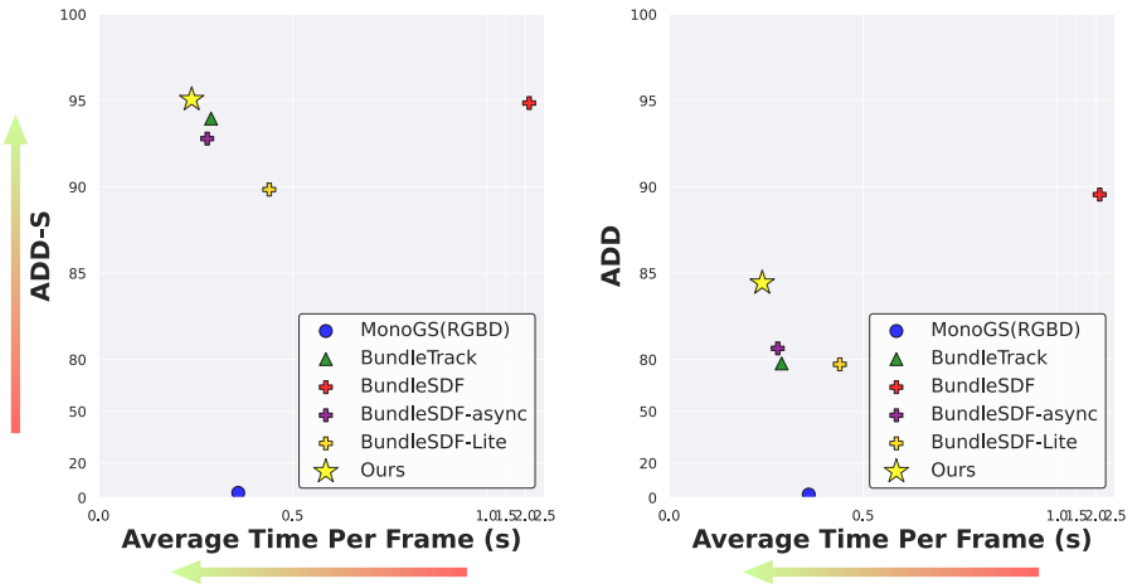


Figure 4. Comparison between temporal efficiency and performance for different approaches on the HO3D dataset. While BundleSDF achieves high performance, it comes at the cost of speed. On the other hand, 6DOPE-GS achieves a favorable tradeoff between speed and performance.

Experiment

Qualitative Comparison (live demo)

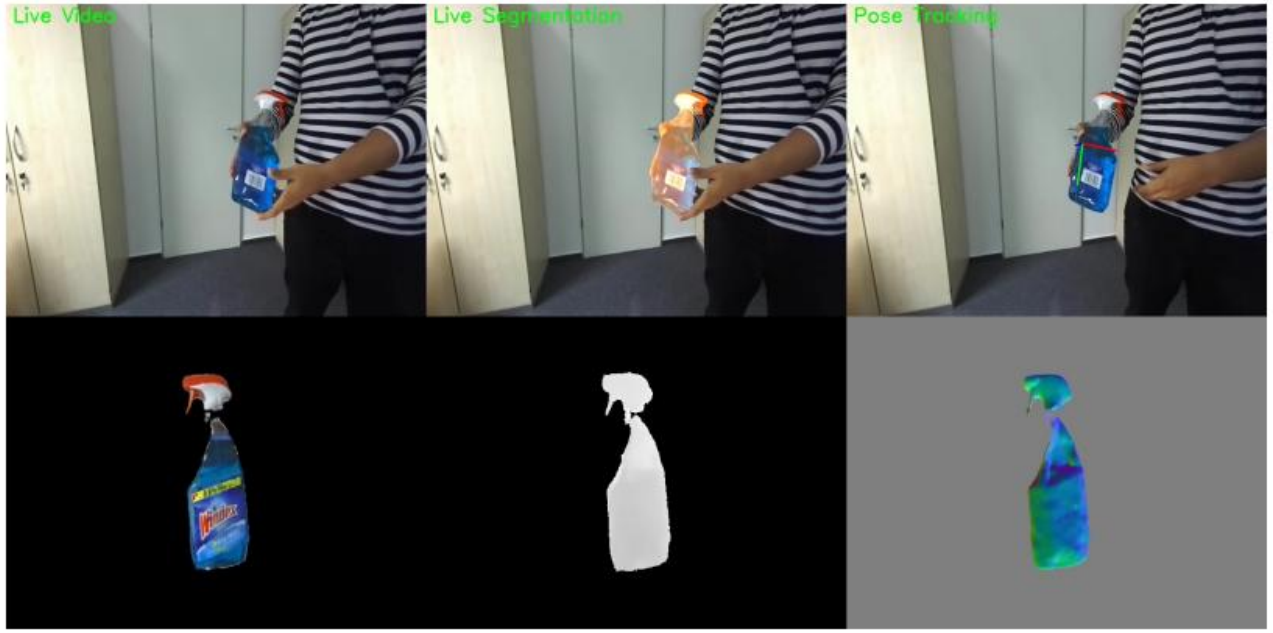


Figure 5. Example of real-time object tracking. **Top row:** Live video, object segmentation results, and pose tracking results. **Bottom row:** Rendered outputs, including color, depth, and surface normals derived from the Gaussian models.

Experiment

Ablation Study

	Method	Pose		Reconstruction
		ADD-S (%) ↑	ADD (%) ↑	CD (cm) ↓
HO3D	Ours (basic)	93.52	80.25	0.44
	Ours w/o KF Selection	94.44	82.40	0.42
	Ours w/o Pruning	92.48	80.87	0.44
	Ours (3DGS)	92.51	79.49	0.47
	Ours (final)	**95.07**	**84.33**	**0.41**
YCBInEOAT	Ours (basic)	92.74	85.15	0.22
	Ours w/o KF Selection	93.03	86.40	0.19
	Ours w/o Pruning	92.64	86.22	0.20
	Ours (3DGS)	91.18	85.29	0.41
	Ours (final)	**93.79**	**87.82**	**0.15**

Table 3. Ablation study of critical design choices

Conclusion

Contribution

- propose a novel method that leverages **2D Gaussian Splatting** for **model-free 6D object pose estimation and reconstruction**.
- propose a **dynamic keyframe selection** approach and a **reconstruction confidence-based filtering** mechanism.
- propose a novel **adaptive Gaussian density control** mechanism based on **opacity percentiles**.

Conclusion

Limitation

- Gaussian rasterization may be **less effective**(Heuristic) **in gradient computations** compared to differentiable ray casting used by NeRFs.
- The **optimized 2D Gaussians** are **not directly integrated** into the **online pose graph optimization**. (**only** utilizes the **pose values**)
- **Dependency** on External Segmentation Models (sam2).
- **Vulnerability** to **Textureless, Symmetric, Transparent** Objects.
- **Lack** of Publicly Available **Source Code**.